



PERCEPTRA

AI-Powered Surveillance & Threat Detection System

PROJECT PROPOSAL

Submitted to: TechRobotics Competition Rwanda

Implementation Period: 3 Months

June 2025

1. Executive Summary

Perceptra (formerly known as Sentry) is an intelligent, AI-powered surveillance and threat detection system designed to revolutionize security monitoring in public and institutional settings. Developed as a submission to the TechRobotics Competition Rwanda, Perceptra leverages computer vision, machine learning, and real-time data analytics to autonomously identify threats, detect anomalous behavior, and generate actionable alerts — reducing reliance on manual surveillance and dramatically improving response times.

The system integrates a smart web-based dashboard with camera feeds, AI inference pipelines, and automated notification modules. In a country like Rwanda, where rapid urban growth, public safety demands, and digital transformation are national priorities, Perceptra presents a timely and locally relevant solution with significant scalability potential across East Africa.

Perceptra aims to make intelligent, automated security accessible and affordable for Rwandan institutions — from schools and hospitals to government buildings and transportation hubs.

2. Problem Statement

Security and surveillance remain major challenges for institutions, public spaces, and communities across Rwanda and the wider East African region. Current security infrastructure in many settings relies heavily on:

- Manual monitoring of CCTV camera feeds, which is prone to human fatigue and attention lapses
- Reactive rather than proactive security responses — threats are identified only after incidents occur
- Limited integration between camera systems and alert/notification infrastructure
- High cost of advanced surveillance systems that are largely imported and inaccessible to local institutions
- Lack of locally developed AI tools suited to the Rwandan and African context

These gaps result in delayed emergency responses, increased vulnerability of public spaces, and over-reliance on human security personnel for tasks that can be automated. There is a clear and urgent need for a cost-effective, locally built, intelligent surveillance solution that can augment or replace legacy systems with AI-driven capabilities.

3. Project Objectives

Primary Objectives

- Develop a fully functional AI-powered surveillance system capable of real-time threat detection and behavioral anomaly identification
- Build and deploy an intuitive, web-based monitoring dashboard accessible to security administrators

- Integrate automated alert systems that notify relevant personnel of detected threats instantly
- Demonstrate a working prototype within the 3-month competition window

Secondary Objectives

- Contribute to Rwanda's Vision 2050 by advancing local AI and robotics capability
- Create a scalable architecture that can be expanded to cover multiple locations and camera nodes
- Establish a framework for future integration with law enforcement and emergency services
- Produce documentation and open resources that support knowledge transfer to local developers

4. Proposed Solution

Perceptra is a multi-layered AI surveillance platform consisting of four integrated components:

4.1 Intelligent Vision Engine

At the core of Perceptra is a computer vision module powered by pre-trained deep learning models (YOLO-based object detection and pose estimation). The engine processes live or recorded video streams to detect people, vehicles, weapons, and suspicious behavioral patterns — including loitering, unauthorized access, and crowd anomalies.

4.2 Smart Web Dashboard

A responsive, browser-based dashboard (built on React and Tailwind) provides security operators with a centralized view of all camera feeds, AI-generated alerts, system status, and historical event logs. The interface is designed for clarity and speed, ensuring operators can act on alerts within seconds.

4.3 Automated Alert & Notification Module

When a threat or anomaly is detected, Perceptra triggers multi-channel alerts — including on-screen notifications, SMS alerts, and (optionally) email notifications — to designated security personnel. Alert severity levels (Low / Medium / High / Critical) allow appropriate escalation protocols to be triggered.

4.4 Data Analytics & Reporting

Perceptra logs all detection events with timestamps and metadata, enabling security administrators to generate incident reports, identify recurring threat patterns, and audit system performance over time.

5. Methodology / Implementation Plan

The project follows an Agile development methodology structured across three one-month sprints, with iterative testing and feedback integration at each phase.

Month 1 — Foundation & Core AI Development

- Requirement gathering and system architecture finalization
- Setup of development environment, version control (GitHub), and project management tools
- Integration and testing of pre-trained computer vision models (YOLO v8)
- Development of video stream processing pipeline
- Initial UI wireframing and front-end skeleton development

Month 2 — Integration & Dashboard Development

- Full dashboard development: live feed display, alert panels, user authentication
- Integration of AI vision engine with the front-end dashboard
- Alert and notification module implementation (in-app, SMS)
- Initial end-to-end testing with simulated and live camera feeds
- Team and feature page content finalization

Month 3 — Testing, Refinement & Presentation

- Comprehensive functional and stress testing
- UI/UX refinements based on internal user testing feedback
- Performance optimization and edge-case handling
- Documentation: technical report, user manual, and competition pitch deck
- Final prototype demonstration and competition submission

6. System Architecture / Technical Design

Perceptra is designed as a lightweight yet powerful system deployable on commodity hardware, making it accessible without requiring expensive infrastructure.

Component	Technology / Detail
Frontend	React, Tailwind css
AI / Vision Engine	Python, OpenCV, YOLOv8 (Ultralytics), PyTorch
Backend API	Python (Flask / FastAPI) for AI inference endpoints
Database	SQLite / PostgreSQL for event logging and audit trails
Alert System	Twilio SMS API, Web Push Notifications

Component	Technology / Detail
Deployment	Local server / Raspberry Pi / Cloud VM (AWS / Azure)
Version Control	GitHub — Perceptra repository
Hardware (Demo)	USB/IP Camera, laptop/mini-PC with GPU or CPU inference

The system follows a client-server architecture where the AI inference engine runs on the server side, processes video frames, and streams detection results and alerts to the web dashboard in near real-time via a REST/WebSocket interface.

Architecture Principle: Perceptra is modular by design — each component (vision engine, dashboard, alert module) can be replaced or upgraded independently, ensuring long-term maintainability.

7. Target Users / Beneficiaries

User / Institution	Use Case
Schools & Universities	Campus security teams monitoring premises for unauthorized access and student safety incidents
Hospitals & Clinics	Facilities requiring 24/7 perimeter monitoring without the cost of large security teams
Government Buildings	Ministries and public offices requiring enhanced access control and threat detection
Transport Hubs	Bus stations and airports needing crowd monitoring and anomaly detection
Shopping Centers & Hotels	Commercial properties requiring real-time surveillance and rapid incident response
Security Companies	Private security firms seeking AI augmentation for their monitoring services

8. Expected Outcomes

Immediate Deliverables (End of 3 Months)

- A fully functional Perceptra prototype with live AI-powered threat detection
- A polished, responsive web dashboard (Home, Features, About, Team, Contact pages)
- Working alert notification system with at least two notification channels
- Comprehensive technical documentation and user manual
- Competition pitch presentation and demo video

Performance Targets

Metric	Target
Detection Accuracy	≥ 85% on standard surveillance benchmark datasets
Alert Response Latency	< 3 seconds from detection to notification
Dashboard Load Time	< 2 seconds on standard internet connection
System Uptime (Demo)	≥ 99% during competition demonstration period
Camera Streams Supported	Up to 4 simultaneous feeds in prototype phase

9. Impact

Social Impact

Perceptra directly contributes to public safety and the well-being of communities in Rwanda by enabling faster, smarter threat detection. Reduced response times to security incidents can save lives and prevent property damage. The system also reduces the cognitive burden on human security personnel, allowing them to focus on higher-order decision-making.

Economic Impact

By offering a locally developed, cost-effective alternative to expensive imported surveillance systems, Perceptra lowers the barrier to AI-powered security for Rwandan institutions. The project also demonstrates the commercial viability of AI product development within Rwanda, creating a template for future local tech ventures.

National & Strategic Impact

- Supports Rwanda's National Digital Economy Policy and Vision 2050 technology pillars
- Builds local AI engineering talent through practical, competitive project development
- Positions Rwandan youth as innovators in the African AI and robotics landscape
- Creates intellectual property owned by Rwandan developers

10. Timeline

Period	Key Activities	Milestone
Month 1 Weeks 1–2	Project setup, architecture design, model selection and integration, version control setup	AI vision engine skeleton complete
Month 1 Weeks 3–4	Video stream pipeline, initial object detection testing, UI wireframes	Working object detection demo

Period	Key Activities	Milestone
Month 2 Weeks 1–2	Dashboard front-end development, API integration, live feed display	Dashboard v1 with live feed
Month 2 Weeks 3–4	Alert module, authentication, end-to-end testing, bug fixing	Full system integration complete
Month 3 Weeks 1–2	User testing, UI/UX refinement, performance optimization	Optimized, tested prototype
Month 3 Weeks 3–4	Documentation, pitch prep, demo video production, final submission	Competition-ready submission

11. Budget

The following is an estimated budget for the 3-month prototype development phase:

Category	Description	Estimated Cost (USD)
Hardware	USB Camera (x2), Mini-PC/Raspberry Pi 5, networking equipment	~\$250
Cloud Services	AWS EC2 / Azure VM (3 months), domain & hosting	~\$90
APIs & Services	Twilio SMS API credits, miscellaneous SaaS tools	~\$30
Development Tools	Software licenses, design tools (Figma, etc.)	~\$20
Competition Costs	Presentation materials, printing, demo setup	~\$50
Contingency (10%)	Unexpected hardware or software costs	~\$44
TOTAL		~\$484

Note: Core software stack (Python, OpenCV, YOLOv8, HTML/CSS/JS) is fully open-source, significantly reducing licensing costs. Hardware costs can be further reduced if the team provides existing devices.

12. Risks and Mitigation

Risk	Mitigation Strategy
Hardware failure / camera issues	Maintain spare hardware; test with recorded video streams as fallback
AI model accuracy below target	Fine-tune on locally relevant datasets; supplement with ensemble methods
Internet connectivity for cloud deployment	Design system for local/offline operation as primary mode
Team member unavailability	Cross-train all members; document all modules for knowledge sharing
Scope creep within 3-month window	Use strict sprint planning and backlog prioritization; lock MVP scope by end of Month 1
Data privacy concerns	Process video locally; no cloud upload of raw footage; comply with Rwandan data protection norms

13. Sustainability and Future Expansion

Post-Competition Roadmap

- Phase 2 (Months 4–6): Expand to support 16+ simultaneous camera streams; add facial recognition module (where legally permitted)
- Phase 3 (Months 7–12): Pilot deployment with a Rwandan institution (school, hospital, or government facility); gather real-world performance data
- Phase 4 (Year 2): Commercialization as a SaaS product for Rwandan and East African security companies; explore government procurement pathways

Revenue & Sustainability Model

- Subscription licensing: Monthly SaaS fees for institutions using the cloud-hosted dashboard
- Hardware bundles: Camera hardware packages pre-configured with Perceptra software
- Maintenance contracts: Annual support and update agreements with institutional clients
- Grants & partnerships: Applications to Rwanda's Innovation Fund, RISA, and international AI development grants

14. Governance and Support

Team Structure

Team Member	Role & Responsibilities
Hashimweyesu Jean de Dieu	Project Lead & AI Engineer — Architecture, ML pipeline, backend

Team Member	Role & Responsibilities
Albert	Frontend Developer — Dashboard UI, web interface
Dorcas	UI/UX Designer — User experience, visual design
Isabelle	Systems Integrator — Hardware, camera integration, testing
Niel	Documentation & Presentation — Technical writing, pitch preparation

Institutional Support

- Competition organizer: TechRobotics Competition Rwanda — providing competition framework, mentorship access, and evaluation criteria
- Technical mentors: To be confirmed upon competition registration
- Development infrastructure: GitHub (version control), shared cloud environment

15. Conclusion / Call to Action

Perceptra represents more than a competition entry — it is a bold demonstration that Rwandan youth can build world-class AI systems that address real, local challenges. Security is a universal need, and AI-powered surveillance that is affordable, locally developed, and ethically designed has the potential to transform how institutions across Rwanda and East Africa protect their people and assets.

In just three months, the Perceptra team will deliver a working, demonstrable system that showcases the depth of local engineering talent and the power of collaborative, mission-driven development. We invite the TechRobotics Competition Rwanda evaluators, mentors, and partners to support, engage with, and invest in the future that Perceptra represents.

We do not just watch — we Perceive, Analyze, and Protect. That is Perceptra.

Submitted by the Perceptra Team

TechRobotics Competition Rwanda
GitHub: github.com/HJeandedieu/SENTRY-AI-Surveillance-System